



Department of Computer Science and Engineering

Lesson Plan: Problem Solving with C [Theory]

Credits: 5

Course code: UE25CS151B

Class #	Unit #	Topics to be covered	% of Portions covered	
			% of syllabus	Cumulative
1	Unit - 1	Introduction to the course Title, Syllabus, Lesson Plan and Clarity on the Evaluation Policy	25%	25%
2		Problem Solving with C in the era of AI, Salient features of C, Programming Environment		
3		Program Development Life cycle of C, Program Structure		
4		Simple Output function, Variables, Data types		
5		Simple Input function, Range of values, Qualifiers		
6		Operators and Expressions		
7		Programming using Variables, Datatypes, Operators, IO functions		
8		Control Structures – Selection Statements, Problem Solving using Selection statements		
9		Control Structures – Looping Statements, Problem Solving using Looping statements		
10		Single Character Input and Output, Problem solving using Control structures and character input/output		
11		Language specifications-Behaviours		
12		Coding standards and Guidelines to avoid different types of errors		
13	Unit - 2	Banana Level Leet Code Challenges	25%	50%
14				
15		Arrays, Problem Solving using Arrays		
16		Pointers, Array of pointers		
17		Functions		
18		Passing an array to a function, Problem Solving using Arrays, functions		
19		Problem Solving using Arrays, functions and array of Pointers		
20		Storage classes		
21		Callbacks		
22		Recursion, Problem solving using recursion		
23		Searching		
24		Searching using array passing to a function		
25		Sorting, Sorting using array passing to a function		
26		Problem Solving using callbacks for searching and Sorting		
27		Strings, Problem Solving using strings		
28		String Manipulation Functions & Errors		

29-30	Unit - 3	Orange Level Challenges in Leet code	25%	75%
31-36	ISA - 1 Week			
37	Unit - 3	Command Line Arguments, Problem Solving using command line arguments		
38		Problem Solving using strings and string functions		
39		Dynamic Memory Allocation and Error Handling		
40		User defined type - Structures, #pragma		
41		Structure Variable, Pointer to structure variable		
42		Passing a structure to a function, Passing pointer to structure to a function, Sample problems		
43		Array of structures, Problem Solving using array of structures and functions		
44		Bitfields		
45		Unions		
46		Enums		
47		Lists - Problem Solving using lists		
48		Stack, Queue - Priority Queue		
49	Unit - 4	FILE IO using redirection	25%	100%
50		File handling functions		
51		Error handling using file handling functions		
52		Perform searching on the contents of the file		
53		Perform sorting on the contents of the file		
54		Header files - Creation and Usage		
55		Comparison of relevant User defined and Built-in functions		
56		Environment variables		
57		Pre-processor directives		
58		Conditional Compilation		
59-62	Jackfruit Level challenges in Leet Code			
63-64	Revision			
65-68	ISA - 2 Week			